

Stroboscopic Eyewear Applied During Warm-Up Does Not Provide Additional Benefits on the Sport-Specific Reaction Speed in Highly Trained Table Tennis Athletes

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Purpose: While long-term training with stroboscopic eyewear suggests performance-enhancing effects on visuomotor abilities, it remains unclear whether a short-term application, for example, during a warm-up, results in immediate performance gains. This study evaluated potential performance-enhancing effects of stroboscopic eyewear applied during warm-up on reaction speed which may provide athletes an edge in visuomotor demanding sports. **Methods:** 28 international-level table tennis athletes participated in this study. Participants performed their individual 10-minute table tennis-specific warm-up under normal visual conditions and with stroboscopic eyewear. Prior to and after the warm-up, the visuomotor reaction time was assessed in a sport-specific reaction test where athletes had to return 30 table tennis balls played by a ball machine at high speed to their backhand side. The reaction time was determined as the interval between ball output and movement onset as triggered by a mechanical switch. Further, the time between ball-table contact and ball-racquet contact (hit time) was analyzed as an indicator of how early the athletes intercepted the ball. **Results:** Reaction time significantly improved following the warm-up ($P < .001$, $\eta_p^2 = .393$). However, there was no additional benefit of the stroboscopic eyewear ($P = .338$, $\eta_p^2 = .034$). No changes after the warm-up were observed for the hit time ($P = .246$, $\eta_p^2 = .055$). **Conclusions:** The results indicate that warm-up facilitated the visuomotor reaction speed; however, stroboscopic eyewear did not provide additional positive effects when compared with to a warm-up under normal visual conditions. While shutter glasses may be useful for training over longer periods, short-term positive effects were not supported in this study.

Keywords: shutter glasses, reaction time, vision, performance, racket sports

With about 80% of sensory input,¹ the visual system is the primary source of information for interaction with the environment. It is particularly important in rapidly changing environments in sport, where visual information must be transferred into a targeted motor response under critical time pressure. As a consequence, athletes develop exceptional visual and visuomotor skills when compared to nonathletes² that are directly related to performance.³ Therefore, the interest in visual training substantially increased.^{3,4}

Over the last years, stroboscopic training has been shown to improve different aspects of visual and visuomotor function including 3-dimensional vision,⁵ motion sensitivity,⁶ and reaction speed.⁷ Further positive effects were observed on sport-specific visuomotor abilities in ice hockey⁸ and badminton.⁹ These effects have been attributed to a more effective processing of visual information following stroboscopic training (see Wilkins and Appelbaum¹⁰ for a review). This was recently supported by neurophysiological analyses indicating that performance gains in reaction time following stroboscopic training were accompanied by a reduced activation latency in the visual motion-sensitive mid-temporal area.¹¹ The results suggest that stroboscopic training induces functional changes in visual systems that mediate performance gains on the behavioral level.

Positive effects were mostly observed in studies applying stroboscopic training over longer periods, typically between 4 and 10 weeks.^{7,10} On an acute level, when performing under

stroboscopic vision, the limitation of visual input delays visual perception in the central nervous system¹² and affects visuomotor performance in reaction tasks and sport-specific assessments.^{13,14} The more challenging environment may not only induce long-term adaptations in the visual system but also benefit visuomotor functions when returning to normal vision. In fact, studies of Bennet et al¹³ and Smith and Mitroff¹⁵ found performance gains in catching and anticipation performance directly after a single session of stroboscopic training. Since Smith and Mitroff¹⁵ applied a short stroboscopic stimulation of only 5 to 7 minutes, it was speculated that using stroboscopic eyewear prior to an event (eg, during warm-up) could lead to immediate benefits on visuomotor performance. In contrast to the assumed functional plasticity following long-term stroboscopic training, immediate effects were hypothesized to be related to higher attention levels following an acute exposure to stroboscopic vision. Since attention is strongly related to visuomotor abilities such as reaction speed,¹⁶ stroboscopic eyewear may be a promising tool to incorporate it into prematch warm-up routines especially in visuomotor demanding sports such as table tennis.

However, as mentioned by Smith and Mitroff,¹⁵ previous research was limited to untrained participants, and need to be confirmed in athletes performing on a high level. Further, the authors used an anticipation task based on a sequence of light-emitting diode lights for both testing and training that does not reflect sport-specific situations. Therefore, it remains unclear whether stroboscopic vision embedded into a sport-specific warm-up benefits sport-specific visuomotor performance in high-level athletes. This

information is of particular interest to athletes and coaches from visuomotor demanding disciplines where immediate performance gains through the warm-up may provide an edge over their opponents.

This study determined the effect of stroboscopic eyewear applied during warm-up on the sport-specific reaction speed in table tennis athletes. 28 international-level table tennis players performed their individual 10-minute table tennis warm-up under stroboscopic and normal visual conditions. Sport-specific reaction tests were performed prior to and after the warm-up. Based on the results of Smith and Mitroff,¹⁵ it was expected that stroboscopic eyewear should provide stronger performance gains after the warm-up when compared to normal vision.

Methods

Based on the results of Smith and Mitroff,¹⁵ effect size calculation revealed a large effect of $d = 0.97$ for difference between groups immediately after the training intervention and a medium effect of $d = 0.51$, 10 minutes after the training which corresponded to an eta square (η^2) of .19 and .06, respectively. Since few minutes were expected between end of the warm-up and starting the posttest in this study, a conservative approach was chosen assuming an effect size of 0.1. Using a 2-factor analysis of variance with the factors TIME (pre and post) and CONDITION (stroboscopic and normal), a P value of .05, and a test power of 0.8, this resulted in a sample size of 20 participants.¹⁷ With 28 participants, this study was sufficiently powered.

Participants and Ethics

28 international-level table tennis athletes between 14 and 37 years of age participated in this study (17 males, 11 females, 25 right-handed, 3 left-handed, age: 21.04 [± 6.4] y). Participants were all members of the national team, participating in international tournaments and selected from junior and adult training groups. On average, participants had 12.89 (± 6.8) years of training experience, a weekly training load of 13.29 (± 4.1) hours, and were ITTF ranked ranked by the International Table Tennis Federation (ITTF) (rank 416 [± 207]). To ensure comparability between conditions (normal and stroboscopic vision), only players who have not used stroboscopic eyewear in their warm-up or training before were selected for this study. Participants were informed about the experimental protocol, and their written consent was obtained. For players under legal age, the consent form was signed by a parent. The study was approved by the local research ethics committee (070/2020) in accordance with the Declaration of Helsinki.

Experimental Protocol

All athletes participated in 2 test sessions of about 30 minutes each.

The 2 sessions assessed the effects of a table tennis-specific warm-up under normal or stroboscopic vision on the visuomotor reaction time. The order of tests was counterbalanced across participants. The time between the 2 tests was at least 24 hours. A familiarization session at least 24 hours prior to the first test ensured that all participants were familiar with the test setup, the task, and the stroboscopic eyewear. The adaptation session had a duration of about 10 minutes and started with a description of the reaction test and stroboscopic eyewear. This was followed by 3 $\times$ 15 balls of practice with the reaction test setup and 2 to 3 minutes of wearing the shutter glasses. The duration corresponded to one interval of stroboscopic exposure in the warm-up. The settings of the shutter

glasses (frequency and duty cycle) were identical to the stroboscopic warm-up condition. Participants did not report experiencing discomfort when using the shutter glasses.

The visuomotor reaction test was designed as a sport-specific table tennis task on a regular table tennis table. A ball robot (Newgy Robo Pong 1050; Newgy Industries Inc) was placed at the end of the table and fully covered to avoid any visual cues indicating ball output. Infrared light barriers were placed directly in front of the robot's ball output to determine stimulus onset (ball output). The ball machine was controlled by a computer running the RoboSoft software package (Newgy Industries Inc). 30 balls subdivided into 2 blocks of 15 trials were played by the robot. The break between the 2 blocks was 30 seconds. The interstimulus interval for the ball output was randomized between 2 and 6 seconds. All balls were played to the backhand side of the player. Ball speed was set that athletes had about 320 milliseconds to hit the ball. This speed setup has previously been used in a similar test and was strongly correlated to laboratory-based reaction time where participants had to react with a button press in response to a radial motion-onset stimulus.¹⁸ Therefore, the test was considered to require the fastest possible reaction speed and represent a valid estimate of the player's reaction performance. Prior to each trial, participants placed their racquet on a mechanical switch positioned at the midline of the table. The mechanical switch was used to determine the reaction time (see below). A target with a radius of 40 cm was placed in the left corner at the opposite side of the player (for left-handed athletes, this setup was mirror-reversed). Athletes were instructed to play the ball to the target to simulate a realistic table tennis action. Next to the contact between the ball and table as well as the anticipated ball-racquet contact, 2 digital microphones were placed to determine the interval between ball-table and ball-racquet contact (hit time). The hit time indicates how early the players intercept the ball on its trajectory after hitting the table. Signals from the light barriers, mechanical switch, and microphones were sampled at 1000 Hz using an actiChamp amplifier (Brain Products) for later analysis. The visuomotor reaction time was defined as the interval between ball output and the participant leaving the mechanical switch. Further, the hit time was determined as the difference between auditory signals from the microphone detecting the ball-table contact and ball-racquet contact, respectively. An illustration of the task setup and the signals recorded during the assessment are presented in Figure 1.

To maximize ecological validity, each athlete did his/her normal warm-up routine prior to a match. The warm-up had a duration of 10 minutes and was performed with a partner playing under normal visual conditions. For the stroboscopic setting, participants used stroboscopic glasses (Senaptec Strobe) that were set to a strobe frequency of 20 Hz at a duty cycle of 50%. These settings were adapted based on previous research with badminton players^{7,11} and accounted for the shorter reaction time in table tennis. Due to the high visual and cognitive load induced by stroboscopic vision, the shutter glasses were used intermittently for 2 to 3 minutes with 1-minute intervals of normal vision in between. After the warm-up, participants were provided 1 minute of time to prepare for the posttest before the task was started. A pause time of 1 minute was chosen to provide players sufficient recovery from the warm-up while avoiding lengthy periods of rest. This setup allowed the investigation of the immediate effects of the stroboscopic warm-up as previously described by Smith and Mitroff.¹⁵ The 1-minute break further provided athletes sufficient time to move without time pressure from the training to the testing table which was in a separate room to avoid disturbance by noise or other players.

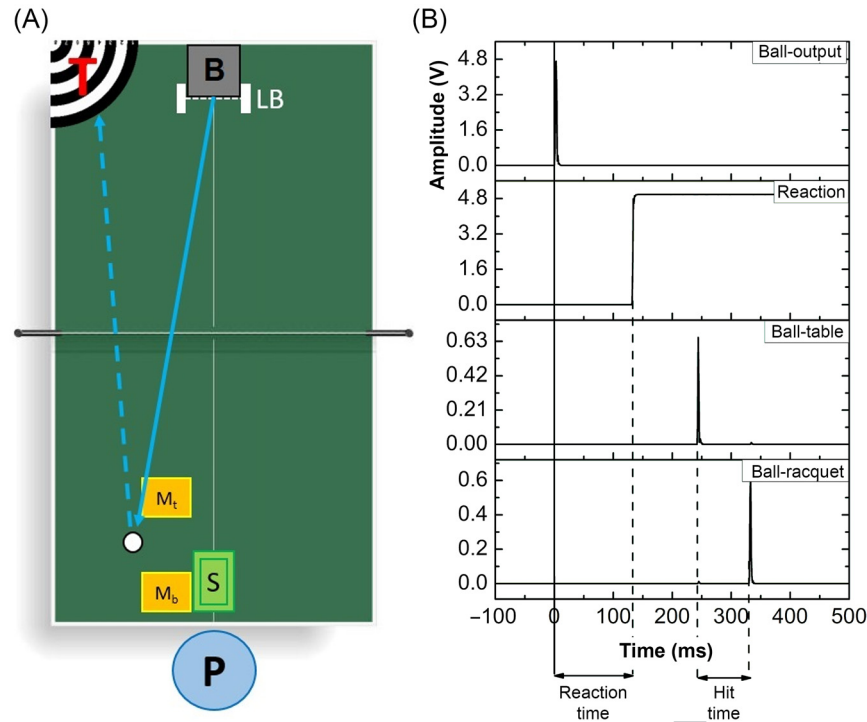


Figure 1 — (A) Illustration of the experimental setup for the table tennis-specific reaction task. B indicates ball machine; LB, light barrier; Mb, microphone capturing the ball-racquet contact; Mt, microphone capturing the ball-table contact; P, participant; S, mechanical switch trigger; T, target. (B) Recording of an exemplary reaction trial by the LBs indicating the ball output (first graph), the S defining the onset of the reaction (second graph), the Mt (third graph), and Mb (fourth graph). The reaction time was defined as the interval between ball output and onset of the S. The hit time reflects the interval between the ball-table contact and ball-racquet contact.

Statistics

All statistical analyses were performed with IBM SPSS (version 28) software. Normal distribution was checked using the Kolmogorov–Smirnov test and confirmed for all parameters. A repeated measures analysis of variance with 2 within-subject factors *CONDITION* (normal and stroboscopic condition) and *TIME* (pretest and posttest) was used to identify main effects of warm-up and differences between normal and stroboscopic warm-up on the visuomotor reaction time and hit time. Correlation analyses on the pretest and posttest results for reaction time and hit time were used to support the reliability of the performance test. Effect sizes were defined based on the thresholds suggested by Cohen¹⁹ and considered small ($\eta_p^2 = .01$), medium ($\eta_p^2 = .06$), or large ($\eta_p^2 = .14$). Significance levels were defined as follows: $*P < .05$, $**P < .01$, and $***P < .001$.

Results

The analysis of variance results are illustrated in Figure 2. A significant main effect was observed for the factor *TIME* ($F_{1,27} = 17.456$, $P < .001$, $\eta_p^2 = .393$) indicating significant improvements in reaction time following warm-up independent of the visual condition. In contrast, the interaction between *TIME* and *CONDITION* did not reach the significance level ($F_{1,27} = 0.950$, $P = .338$, $\eta_p^2 = .033$). For the hit time reflecting the interval between ball-table and ball-racquet contact, neither a main effect for *TIME* ($F_{1,24} = 1.412$, $P = .246$, $\eta_p^2 = .055$) nor an interaction effect for *CONDITION* ($F_{1,24} = 0.070$, $P = .793$, $\eta_p^2 = .002$) was observed.

Pearson correlation analyses revealed strong correlations between the pretest and posttest reaction time in both the normal ($r_{28} = .92$, $P < .001$) and stroboscopic vision ($r_{28} = .90$, $P < .001$) condition. This also applied to the hit time (normal: $r_{25} = .93$, $P < .001$, stroboscopic: $r_{25} = .89$, $P < .001$). Figure 3 shows the correlation results. A control analysis investigating the correlation between both pretests which can be considered as an indicator of between-day reliability likewise revealed a highly significant correlation ($r_{25} = .79$, $P < .001$).

Discussion

While previous studies suggest positive effects of long-term stroboscopic training on visuomotor performance, research on short-term effects is scarce. Few studies investigated immediate effects but were limited by nonathlete participants and generic testing and training tasks. Therefore, it remained unclear whether stroboscopic eyewear used in a sport-specific warm-up improves visuomotor performance in professional sports. International-level table tennis athletes completed a sport-specific visuomotor reaction task prior to and after a regular warm-up performed under normal or stroboscopic visual conditions. While the warm-up improved visuomotor reaction speed, the results did not reveal an additional benefit of incorporating stroboscopic vision into the warm-up routine. While previous research supported the positive effects of long-term stroboscopic training interventions over several weeks,^{7,10} this study did not provide quantitative evidence supporting its benefits on visuomotor reaction performance for athletes when used during warm-up.

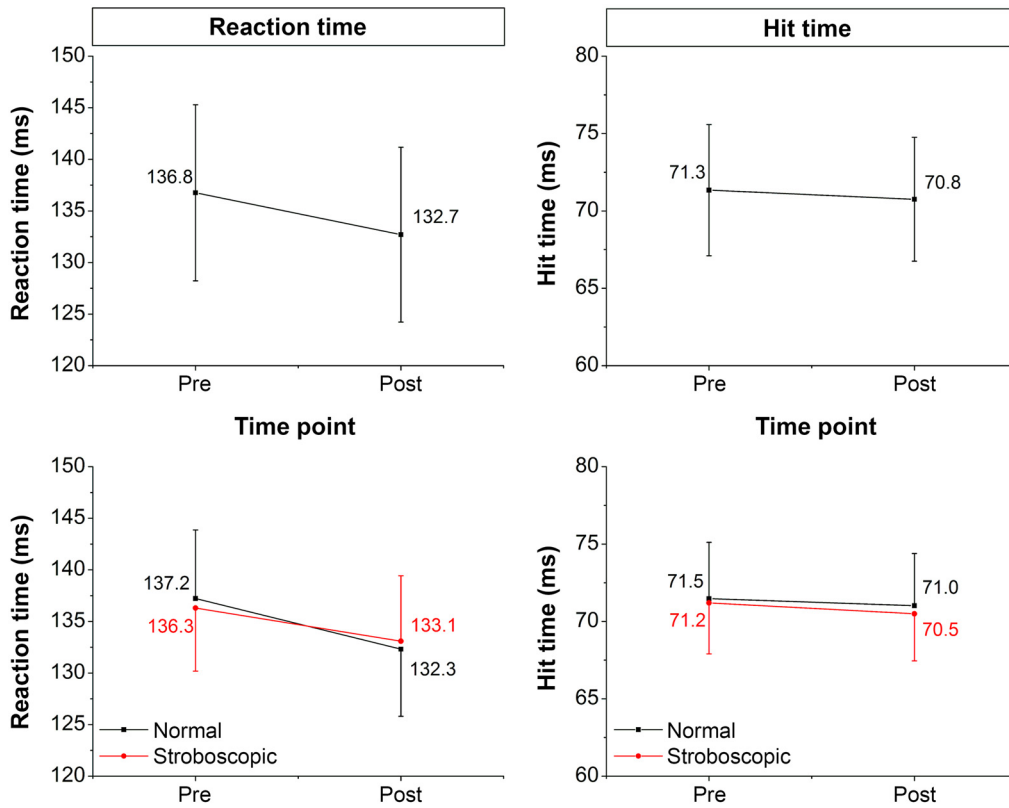


Figure 2 — ANOVA results on the main effect for time (pre and post) and the interaction effect between the factors time (pre and post) and condition (stroboscopic and normal vision) indicated for the reaction time and hit time. Error bars indicate 95% confidence intervals. ANOVA indicates analysis of variance.

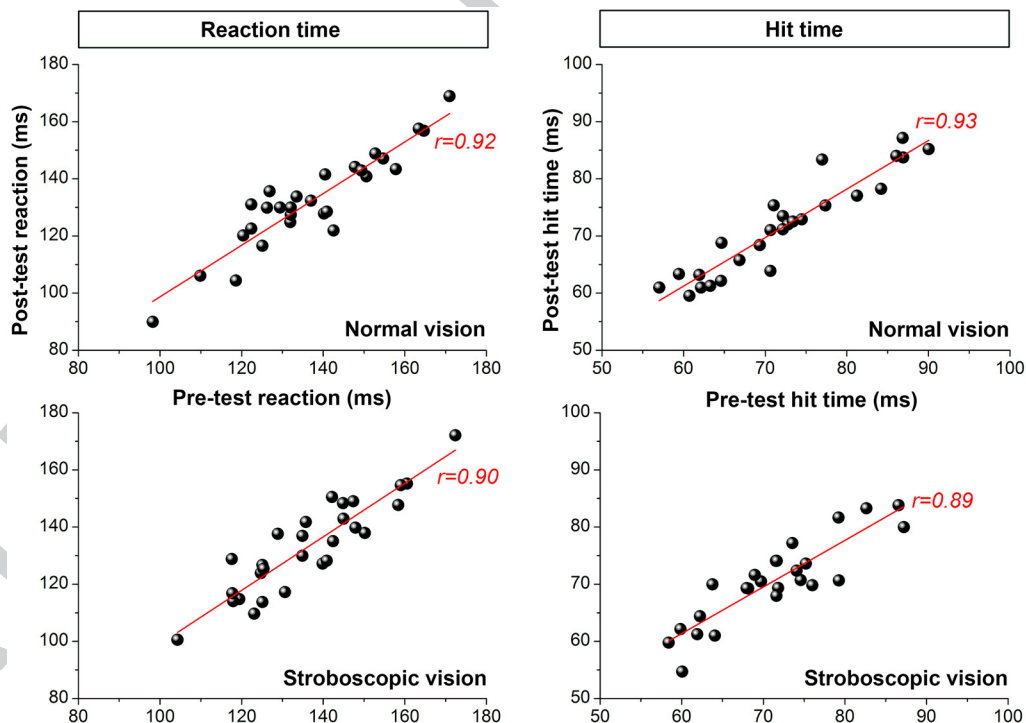


Figure 3 — Correlation analyses between pretest and posttest reaction time and hit time under normal and stroboscopic conditions.

Independent of the visual condition, the table tennis warm-up significantly improved visuomotor reaction performance as reflected by faster reaction times. These findings add further support to the well-established beneficial effects of warm-up on various aspects of sport performance.²⁰ Concerning the reaction time investigated in this study, the results are further in line with recent studies in the field of football.^{21,22} Following a match-specific warm-up, goalkeepers were observed to achieve significantly faster reaction speeds.²¹ Importantly, this was only confirmed after a sport-specific warm-up while a nonspecific preparation not considering football-specific characteristics did not change the reaction speed.

In contrast to reaction time, the hit time reflecting the interval between ball-table and ball-racquet contact did not significantly change after the warm-up. This suggests that although athletes reacted faster, they did not intercept the ball earlier on its trajectory after touching the table. The unchanged hit time indicates that athletes may have used the additional time for movement execution to achieve better outcomes according to the speed-accuracy trade-off.²³

Although warm-up positively affected reaction speed, there was no additional benefit of incorporating stroboscopic eyewear into the procedure. These findings contradict our hypothesis and are in contrast to the study of Smith and Mitroff¹⁵ suggesting strong positive effects of short-term stroboscopic interventions on anticipation. This discrepancy can best be explained by differences in the warm-up procedure and sample characteristics. While Smith and Mitroff¹⁵ used a generic anticipation task with nonathletes, this study applied the player's individual warm-up strategy in high-level table tennis to maximize ecological validity. Participants in the study of Smith and Mitroff¹⁵ were unfamiliar with the task and trained it either under normal or stroboscopic conditions. In contrast, athletes in this study were highly accustomed to their individual pregame warm-up routine. Due to the well-established importance of preperformance routines in elite sports (see Rupprecht et al.²⁴ for a meta-analysis), stroboscopic eyewear may interfere with the athlete's warm-up routine. Consequently, potential performance gains induced by the stroboscopic stimulus may have canceled out with negative effects due to interfering with the warm-up routine.

From a methodological perspective, it may be argued that the task was not selective for reaction speed and athletes were not forced to respond as fast as possible. However, since reaction time improved after warm-up in both conditions, it can be assumed that faster reactions were beneficial for task performance. Further, the strong correlation between reaction time in a similar table tennis task and a standardized laboratory-based reaction test suggest that the test applied in this study primarily assesses the athlete's reaction time.¹⁸ Potential masking effects where the effects of a warm-up in general may have outweighed the additional effect of stroboscopic eyewear are unlikely due to the very small effect size in the interaction effect. Finally, the very strong correlations between the pretest and posttest reaction performance in both conditions ($r > .89$) as well as between both pretests that were performed under identical conditions but on different days ($r = .79$) support the reliability of the sport-specific reaction test and consistent changes from pretest to posttest.

Practical Application

In contrast to positive long-term effects of stroboscopic training, the results do not support a positive effect when incorporating stroboscopic eyewear during a sport-specific warm-up. This argues against previously hypothesized positive effects on attention that may benefit performance in visuomotor tasks. Therefore, athletes

and coaches can be advised that shutter glasses can be a promising training tool when applied over longer training periods especially since positive training effects are suggested to last for at least several weeks.⁷ However, immediately prior to the match, athletes should follow their regular warm-up routine without stroboscopic eyewear. Nonetheless, since there was no detrimental effect of shutter glasses on performance, athletes who are already using stroboscopic eyewear for their warm-up do not have to expect negative effects on their performance.

Limitations and Future Directions

This study used a pretest and posttest design where the pretest could already be considered a warm-up for the posttest. This may have interfered with the sport-specific warm-up and affected the results. Future research using a passive control condition (no warm-up) should control for this possibility. Moreover, follow-up studies may investigate the physiological demands associated with the warm-up and the reaction test to identify potential effects of physical activity level on the reaction performance. Finally, a prerequisite for the players to participate in this study was that they have not used stroboscopic eyewear in their training or warm-up before. While this ensured a comparable baseline level, it may be argued that, although there was a familiarization session, initial negative effects of the stroboscopic warm-up due to unfamiliarity with the technology may have negatively affected the results. Future research should monitor the participant's subjective task complexity and perceived workload, as well as the effects of stroboscopic warm-up over longer time periods, to elaborate on this hypothesis. This may be combined with an individual adjustment of the stroboscopic settings (frequency and duty cycle) to adjust the task difficulty and account for interindividual differences in performance.

Conclusions

The combined pattern of result did not support the benefit of incorporating stroboscopic eyewear into a pre-training or pre-competition warm-up in visuomotor demanding disciplines such as table tennis. While stroboscopic training may be beneficial to improve visual and visuomotor function over longer training periods, immediate effects were not observed. Therefore, there is currently no quantitative evidence for using stroboscopic eyewear during sport-specific warm-up routines to improve visuomotor reaction performance.

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